Research

A:Geometry - Geom - SpaceClaim

File Sketch Design Display Assembly Measure Facets Repair Prepare Workbench Detail Sheet Metal Tools KeyShot Ansys Motion

Paste | Home Spin Plan View Pan Zoom | Sketch Mode | Select Pull Move Fill | Combine Split Body Split Project | Shell Offset Mirror | Equation Cylinder Sphere | Blocks Script

Clipboard Orient Mode Edit Intersect Create Body Record

Structure

- 020420516*
  - Facets
  - Plane
  - Plane
  - Plane

Structure Layers Selection Groups Views

Options - Selection

Properties

Properties Appearance

Ansys 2024 R1 TEACHING

Geom*

Ready

A:Geometry - Geom - SpaceClaim

File | Sketch | Design | Display | Assembly | Measure | Facets | Repair | Prepare | Workbench | Detail | Sheet Metal | Tools | KeyShot | Ansys Motion

Home | Plan View | Orient

Auto Skin | Skin Surface | Extract Curves | Orient Facets | Fit Spline | Vectorize Image — Reverse Engineering

Standard Hole | Identify Holes | Move Body | Create Workpiece | Toolpath | Relief | Unroll — Manufacturing

Structure

- 020420516*
  - Facets
  - Plane
  - Plane
  - Plane

Structure | Layers | Selection | Groups | Views

Options

Options

Samples 50

Full preview

Properties

Properties | Appearance

Create a boundary for the skin: Click to add control points, double-click to add vertices

Ansys 2024 R1 TEACHING

Geom*

Create a boundary for the skin: Click to add control points, double-click to add vertices

A:Geometry - Geom - SpaceClaim

File | Sketch | Design | Display | Assembly | Measure | Facets | Repair | Prepare | Workbench | Detail | Sheet Metal | Tools | KeyShot | Ansys Motion

Home | Plan View | Orient

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Standard Hole | Identify Holes | Move Body | Create Workpiece | Toolpath | Relief | Unroll — Manufacturing

Structure

- 020420516*
  - Surface
  - Facets
  - Plane
  - Plane
  - Plane

Structure | Layers | Selection | Groups | Views

Options

General

Keep bend lines

Properties

Properties | Appearance

Ansys 2024 R1 TEACHING

Geom*

Ready

Face

A:Geometry - Geom - SpaceClaim

File | Sketch | Design | Display | Assembly | Measure | Facets | Repair | Prepare | Workbench | Detail | Sheet Metal | Tools | KeyShot | Ansys Motion

Home | Plan View | Orient | Convert | Square-Up | Identify | Import | Sketch | Select | Pull | Move | Edit | Junction | Edge Relief | Corner Relief | Modify | Forms | Split | Marker | Tab | Bead | Double wall | Hinge | Bend | Gusset | Flange | Create | Unfold | Fold Walls | Flat | Highlight | Show

Structure

- 020420516*
  - Surface
  - Unrolled
  - Facets
  - Plane
  - Plane
  - Plane

Structure | Layers | Selection | Groups | Views

Options - Move

General

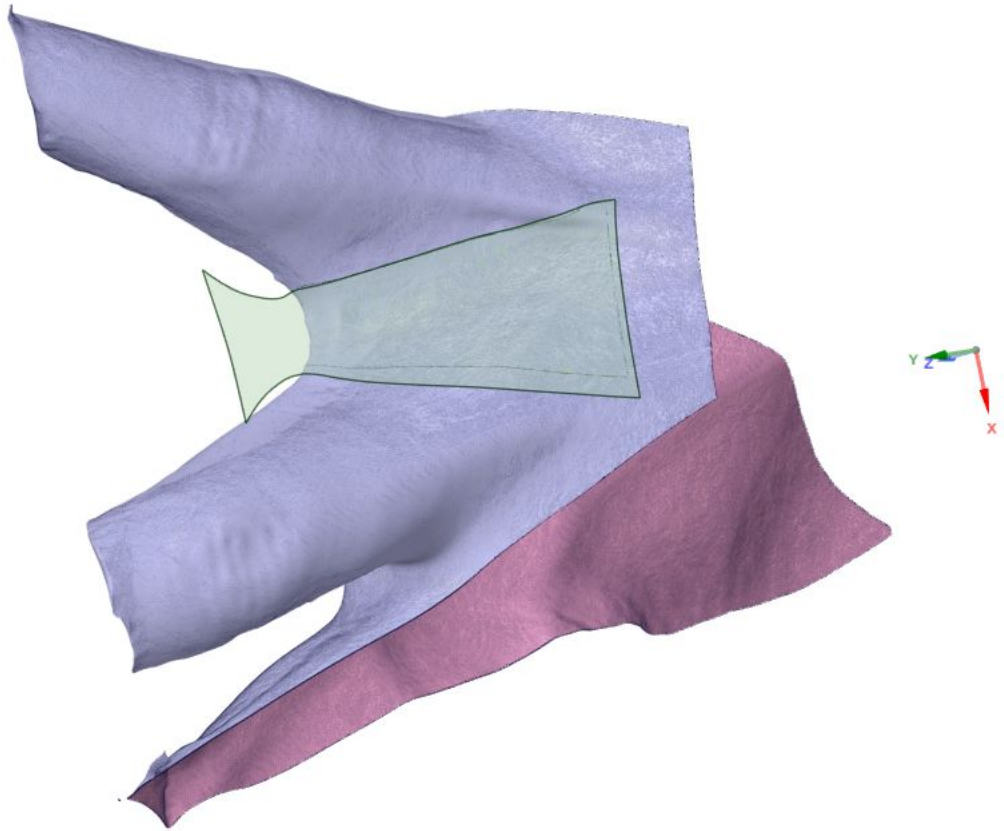
Create patterns
Maintain orientation
Detach first
Maintain sketch connectivity
Remember orientation | Default

Properties

Properties | Appearance

Click an object. Double-click to select a chain or loop. Triple-click to select a solid.

Ansys 2024 R1 TEACHING



Y Z X

Geom*

Click an object. Double-click to select a chain or loop. Triple-click to select a solid.

B:SpaceclaimAndDesign - Geom-1 - SpaceClaim

File Sketch Design Display Assembly Measure Facets Repair Prepare Workbench Detail Sheet Metal Tools KeyShot Ansys Motion

Home Plan View Orient | Measure Mass Properties Check Geometry Inspect | Curves Volumes Interference | Normal Grid Curvature Dihedral Draft Stripes Deviation Quality

Measure (E)
Click objects to measure. The properties can be copied to the clipboard for pasting.
Press F1 for more help, or F3 for a video.

Structure
- Geom-1*
  - Surface
  - Unrolled
  - Facets
  - Plane
  - Plane
  - Plane
  - *Curves*

Structure Layers Selection Groups Views

Options - Pull
General
Add Cut No merge

Properties

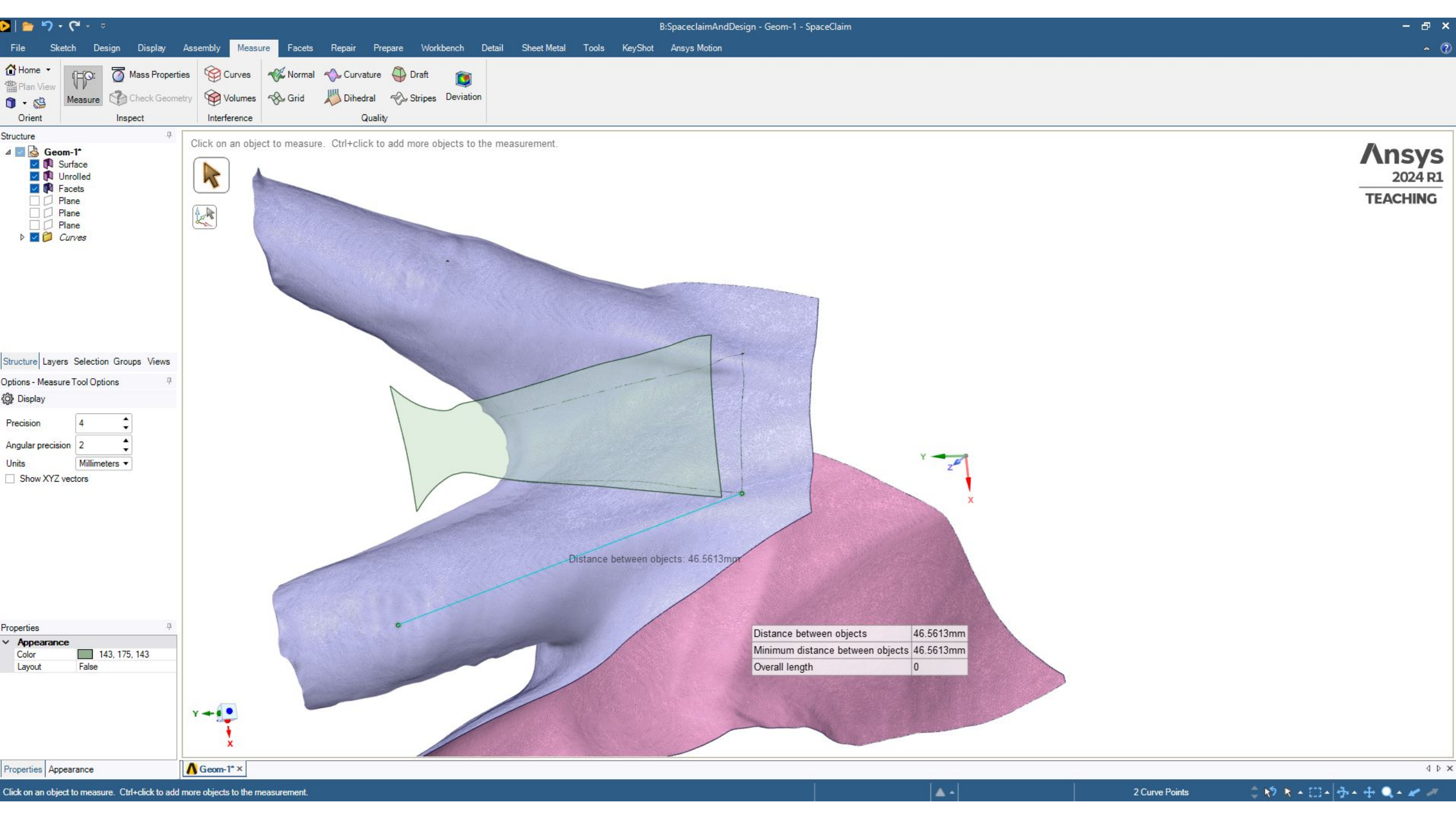
Properties Appearance

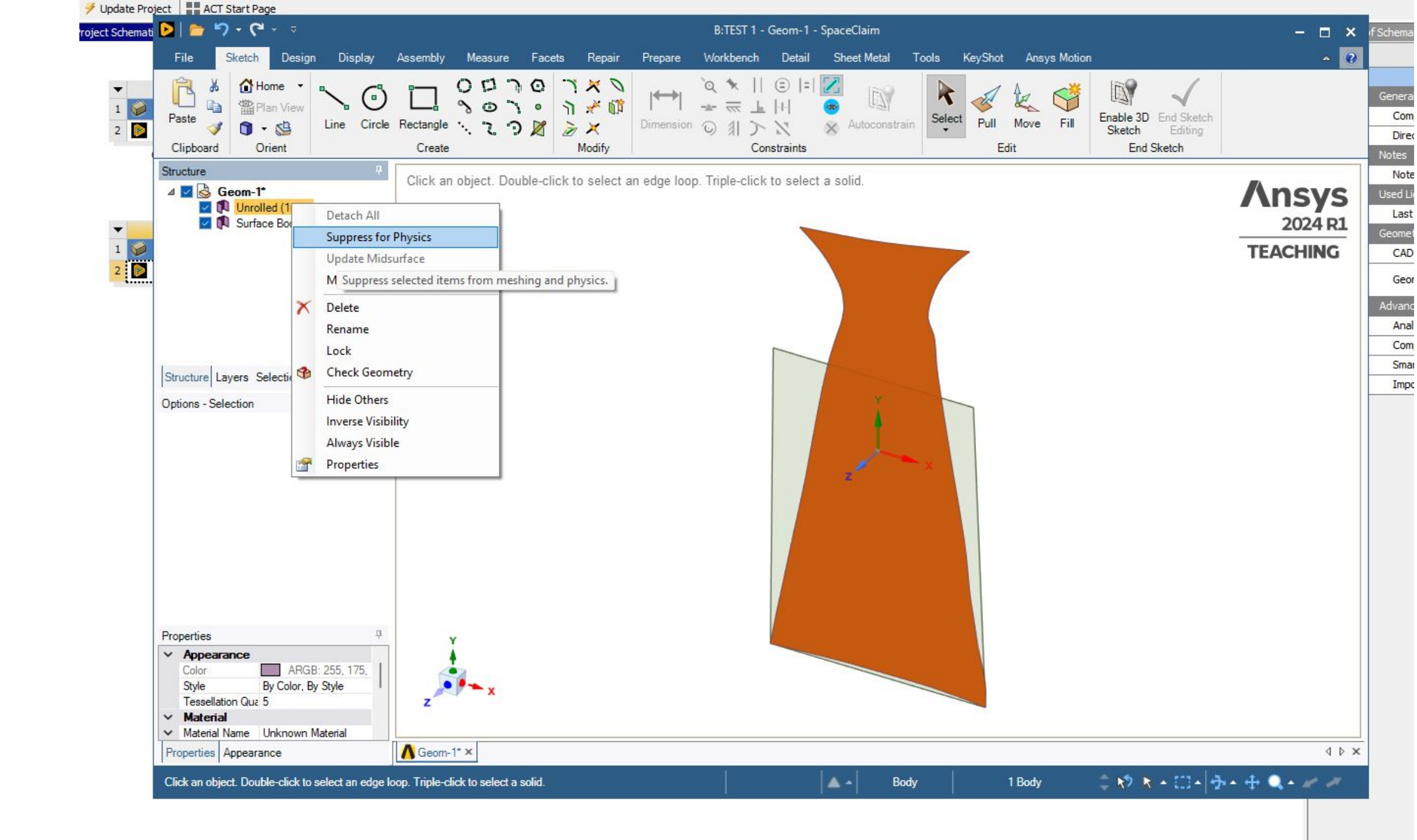
Ansys 2024 R1 TEACHING

Draw dots on the points
Measure the length between the points
38.89 mm upper
46.56 mm down

Geom-1* ×

Select and drag a face to offset it. Select and drag an edge to round it.


B:SpaceclaimAndDesign - Geom-1 - SpaceClaim
File Sketch Design Display Assembly Measure Facets Repair Prepare Workbench Detail Sheet Metal Tools KeyShot Ansys Motion
Home Plan View Orient
Measure Mass Properties Check Geometry Inspect
Curves Volumes Interference
Normal Curvature Draft Grid Dihedral Stripes Deviation Quality
Structure
Geom-1*
Surface
Unrolled
Facets
Plane
Plane
Plane
Curves
Structure Layers Selection Groups Views
Options - Measure Tool Options
Display
Precision 4
Angular precision 2
Units Millimeters
Show XYZ vectors
Properties
Appearance
Color 143, 175, 143
Layout False
Properties Appearance
Click on an object to measure. Ctrl+click to add more objects to the measurement.
Ansys 2024 R1 TEACHING
Distance between objects: 46.5613mm
Distance between objects 46.5613mm
Minimum distance between objects 46.5613mm
Overall length 0
Geom-1*
Click on an object to measure. Ctrl+click to add more objects to the measurement.
2 Curve Points


Update Project
ACT Start Page
B:TEST 1 - Geom-1 - SpaceClaim
File
Sketch
Design
Display
Assembly
Measure
Facets
Repair
Prepare
Workbench
Detail
Sheet Metal
Tools
KeyShot
Ansys Motion
Paste
Clipboard
Home
Plan View
Orient
Line
Circle
Rectangle
Create
Modify
Dimension
Autoconstrain
Constraints
Select
Pull
Move
Fill
Edit
Enable 3D Sketch
End Sketch Editing
End Sketch
Structure
Geom-1*
Unrolled (1
Surface Bo
Detach All
Suppress for Physics
Update Midsurface
M
Suppress selected items from meshing and physics.
Delete
Rename
Lock
Check Geometry
Hide Others
Inverse Visibility
Always Visible
Properties
Structure
Layers
Selecti
Options - Selection
Click an object. Double-click to select an edge loop. Triple-click to select a solid.
Ansys
2024 R1
TEACHING
Properties
Appearance
Color
ARGB: 255, 175,
Style
By Color, By Style
Tessellation Qua
5
Material
Material Name
Unknown Material
Properties
Appearance
Geom-1*
Click an object. Double-click to select an edge loop. Triple-click to select a solid.
Body
1 Body

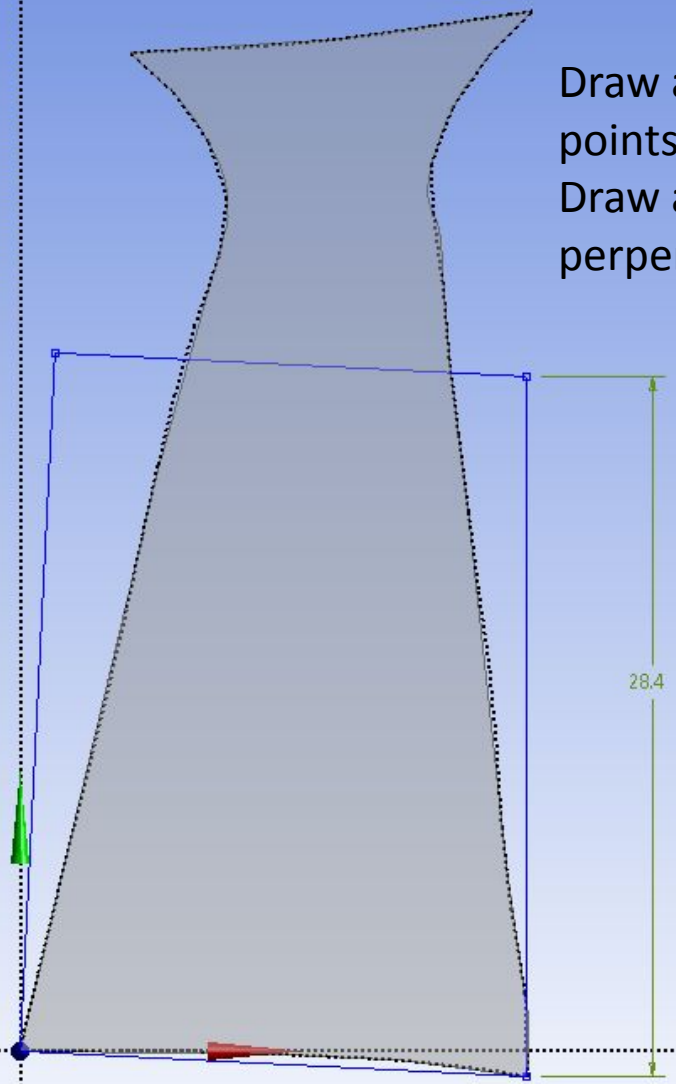
- After measure, delete everything but unrolled surface
- And move to the XY plane
- Save as a Parasolid(X_T or X_ B) -> this shortens the time to import model from Spaceclaim to DesignModeler
- -> Why do I use DesignModeler instead of spaceclaim???

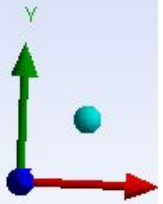

Unsaved Project - Workbench
File View Tools Units optiSLang Extensions Jobs Help
Project
Import... Reconnect Refresh Project Update Project ACT Start Page
Toolbox
Analysis Systems
Coupled Field Harmonic
Coupled Field Modal
Coupled Field Static
Coupled Field Transient
Eigenvalue Buckling
Electric
Explicit Dynamics
Fluid Flow (CFX)
Fluid Flow (Fluent with Fluent Meshing)
Fluid Flow (Fluent)
Fluid Flow (Polyflow)
Harmonic Acoustics
Harmonic Response
Hydrodynamic Diffraction
Hydrodynamic Response
LS-DYNA
LS-DYNA Restart
Magnetostatic
Modal
Modal Acoustics
Motion
Random Vibration
Response Spectrum
Rigid Dynamics
Static Acoustics
Static Structural
Steady-State Thermal
Structural Optimization
Substructure Generation
Thermal-Electric
Transient Structural
Transient Thermal
Turbomachinery Fluid Flow
Component Systems
Custom Systems
Design Exploration
optiSLang
optiSLang Integrations
ACT
View All / Customize...
Project Schematic
A
1 Geometry
2 Geometry
Spaceclaim
B
1 Geometry
2 Geometry
SpaceclaimAndDesign
It takes forever to import model in Designmodeler
Double-click component to edit.
Job Monitor... No DPS Connection No HPC Platform Services Connection Show Progress Show 1 Messages

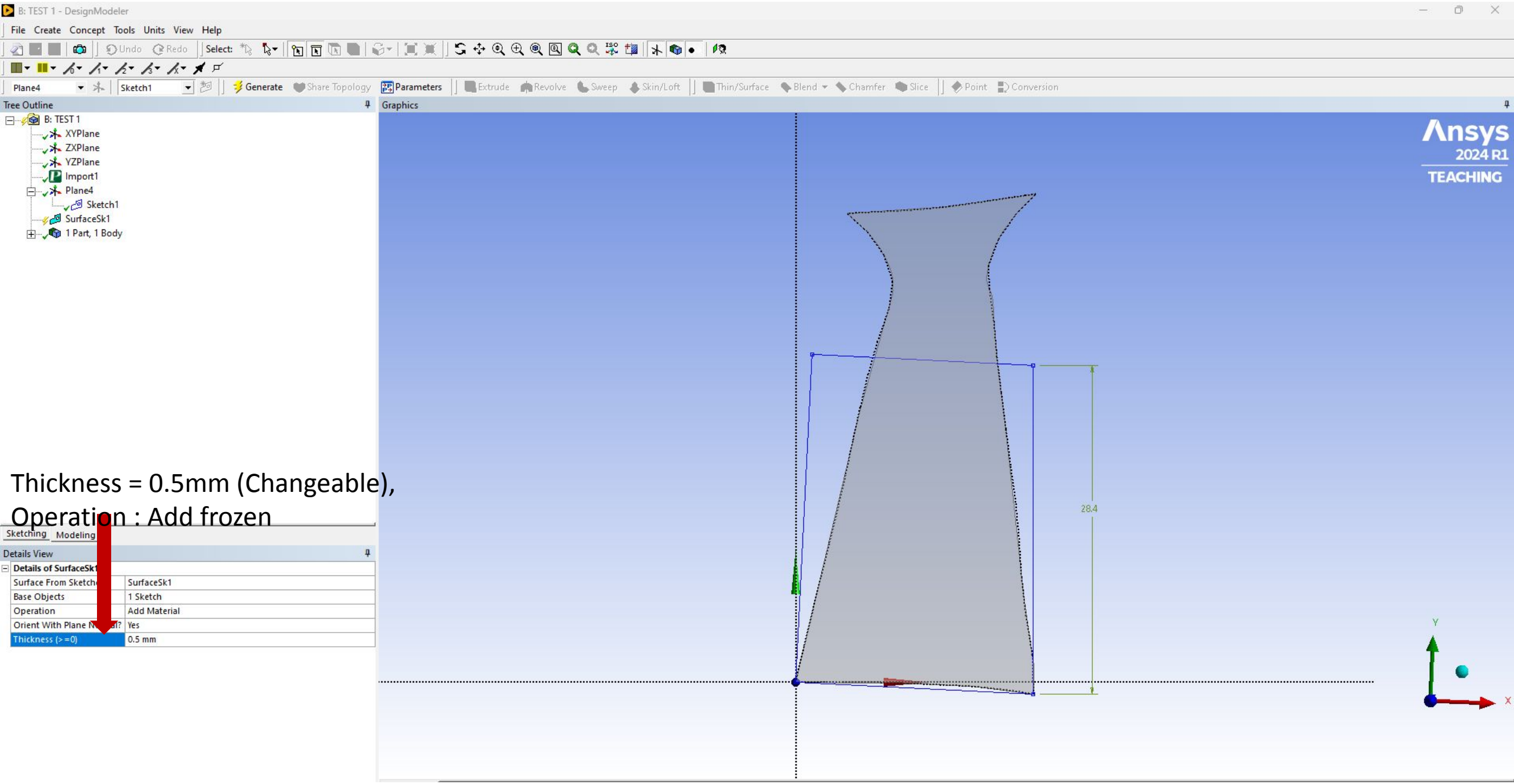
Graphics

The average of the lengths*2/3 -> (38.9 + 46.5)/2*2/3 = 28.4

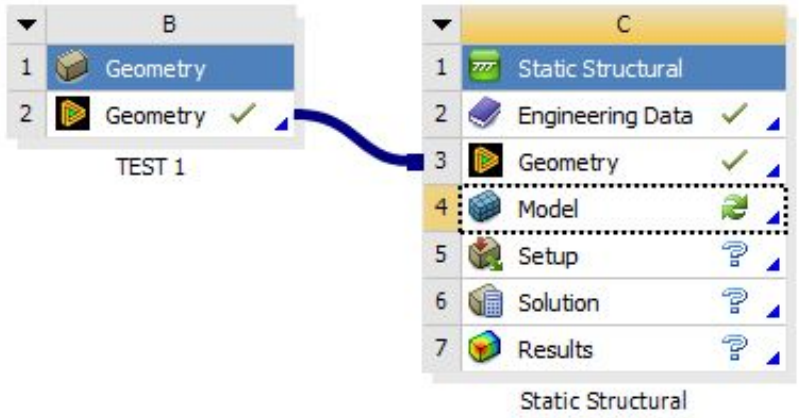
Draw a single line which connects two points on the bottom
Draw a rectangular, make sure the lines are perpendicular and parallel






B: TEST 1 - DesignModeler
File Create Concept Tools Units View Help
Undo Redo Select:
Plane4 Sketch1 Generate Share Topology Parameters Extrude Revolve Sweep Skin/Loft Thin/Surface Blend Chamfer Slice Point Conversion
Tree Outline
B: TEST 1
XYPlane
ZXPlane
YZPlane
Import1
Plane4
Sketch1
SurfaceSk1
1 Part, 1 Body
Graphics
Ansys
2024 R1
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28.4
Y
X
Thickness = 0.5mm (Changeable),
Operation : Add frozen
Sketching Modeling
Details View
Details of SurfaceSk1
Surface From Sketch SurfaceSk1
Base Objects 1 Sketch
Operation Add Material
Orient With Plane Normal? Yes
Thickness (>=0) 0.5 mm

Start Analysis



Properties of Schematic B2: Geometry

	A	B
1	Property	Value
2	General	
3	Component ID	Geometry 2
4	Directory Name	Geom-1
5	Notes	
6	Notes	
7	Used Licenses	
8	Last Update Used Licenses	
9	Geometry Source	
10	Geometry File Name	
11	Advanced Geometry Options	
12	Analysis Type	2D
13	Compare Parts On Update	No

Skin Properties

Properties of Outline Row 3: Skin of hand

	A	B	C	D	E
1	Property	Value	Unit		
2	Material Field Variables	Table			
3	Density	1100	kg m^-3		
4	Isotropic Secant Coefficient of Thermal Expansion				
5	Coefficient of Thermal Expansion	0.0001	C^-1		
6	Isotropic Elasticity				
7	Derive from	Young's Modulus and Poisson...			
8	Young's Modulus	8E+05	Pa		
9	Poisson's Ratio	0.47			
10	Bulk Modulus	4.4444E+06	Pa		
11	Shear Modulus	2.7211E+05	Pa		

Make sure choosing material as skin on Mechanical

Context B : Static Structural - Mechanical [Ansys Mechanical Enterprise]

File Home Geometry Display Selection Automation Add-ons Learning and Support

Quick Launch

Create | Hide Bodies | Show Bodies | Hide Others | Suppress Bodies | Unsuppress Bodies | Suppress Others — Named Selections

Adjacent | Limits | Connection | Instances — Extend To

Mesh by Id | Location | Size | Invert | Common Edges | Cylindrical Faces | Shared Topology | Same Material | Grow Element — Select

Bodies | Edges | Faces | Vertices | Shared — Convert To

Start | Previous | Next | First | Last | Reselect All — Walk

GoTo — Tree Select

Scoping Wizard — Tools

Outline

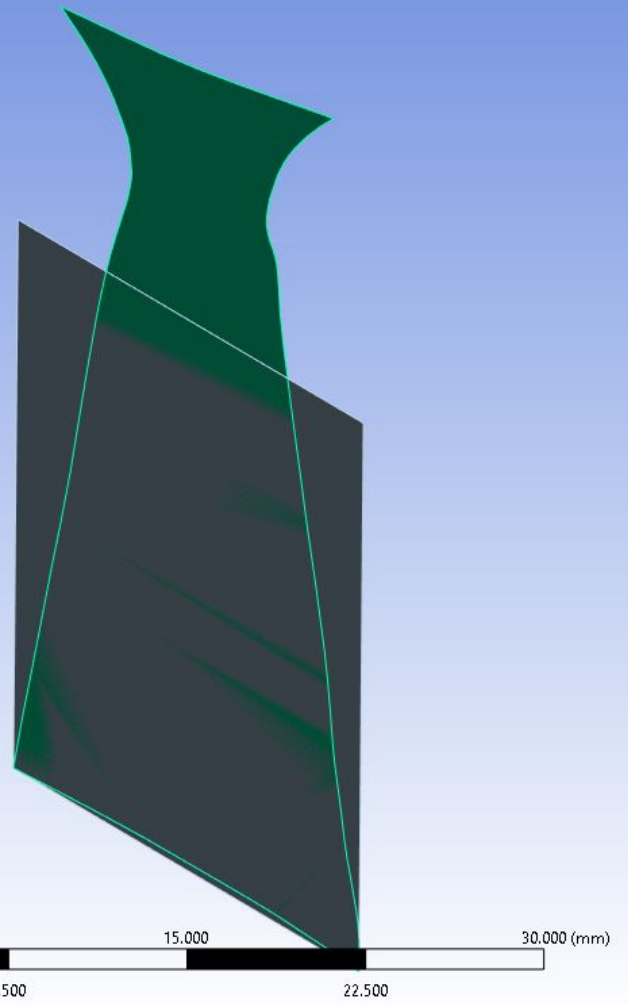
Name | Search Outline

- Project*
  - Model (B4)
    - Geometry Imports
    - Geometry
      - Unrolled
      - Su
    - Materia
    - Coordin
    - Connec
    - Mesh
    - Named
    - Static
      - An
      - Sc

Select | Mode | Clipboard | [Empty] | Extend | Select By | Convert

Geom\Unrolled
2/18/2025 5:00 PM

Ansy
2024 R
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0.000 15.000 30.000 (mm)
7.500 22.500

Y
Z X

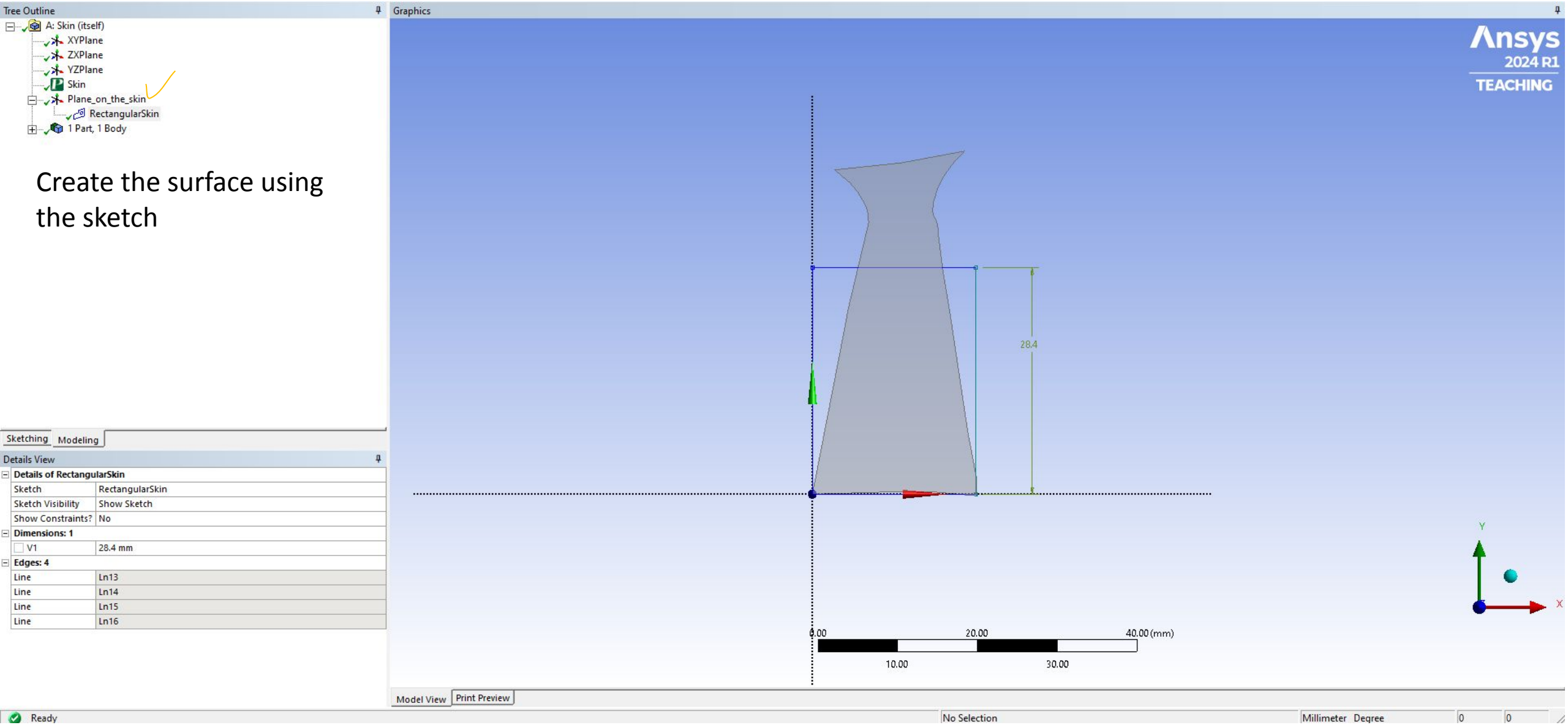
Insert
Update
Generate Mesh
Preview
Freeze Mesh
Hide Body
Hide All Other Bodies
Rename F2
Group Ctrl+G
Suppress Body
Suppress All Other Bodies
Create Named Selection
Create Mate
Create Mate
Transform Pa
Delete Part(s
Export...
Update Selected Parts
Clear Generated Data

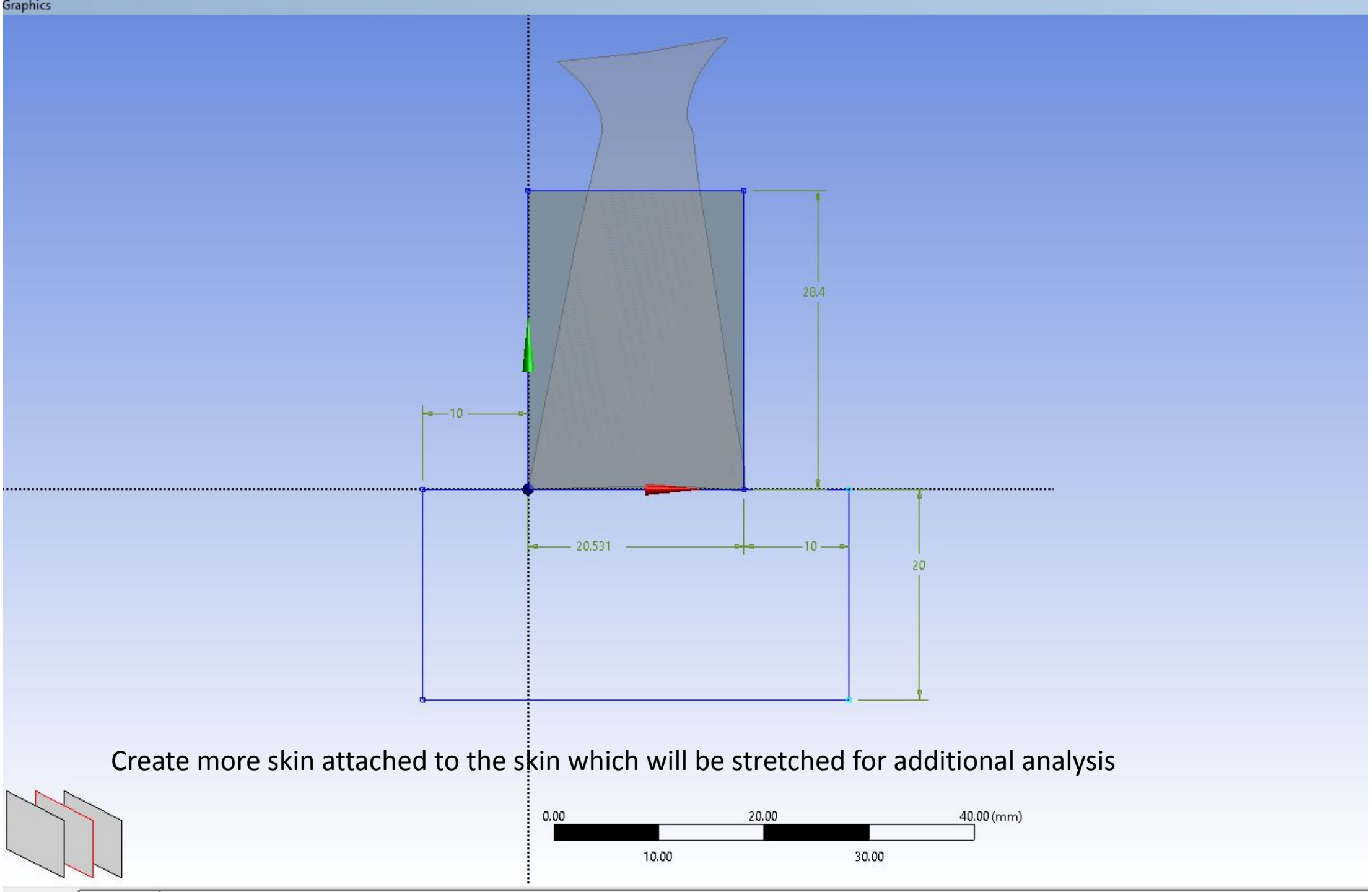
Create Named Selection
Create a Named Selection for the selected geometry entities from the Geometry object. You can specify a name for the selection.
Press F1 for help.

Details of "Unrolled"

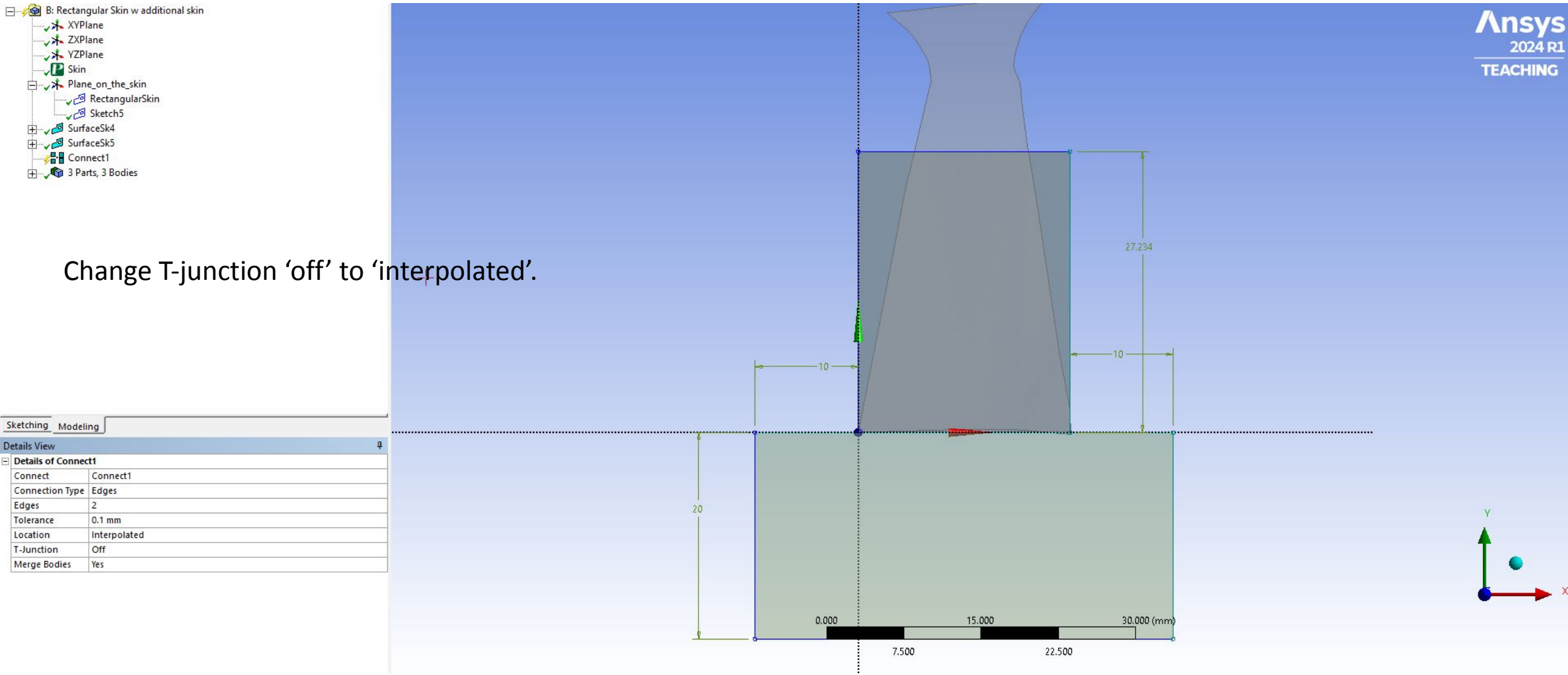
Graphics Properties	
Definition	
Suppressed	No
Dimension	3D
Model Type	Shell
Stiffness Behavior	Flexible
Stiffness Option	Membrane and Bending
Coordinate System	Default Coordinate Sy...
Reference Temperature	By Environment
Thickness	0.5 mm
Thickness Mode	Refresh on Update
Offset Type	Middle
Treatment	None
Material	
Assignment	Structural Steel
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	

Second Trial





Create more skin attached to the skin which will be stretched for additional analysis



Change T-junction ‘off’ to ‘interpolated’.

Sketching | Modeling

Details View

Details of Connect1	
Connect	Connect1
Connection Type	Edges
Edges	2
Tolerance	0.1 mm
Location	Interpolated
T-Junction	Off
Merge Bodies	Yes

Result